Presentation 4

Link to Presentation 4 Slides
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/MenuRepetitionsandSuccessfulCallAnalysis_18Nov_WillVinceYiHyun.pdf

Link to Code
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Menu%20redundancy%20analysis/MainMenuClusterAnalysis_18Nov_YiHyun.ipynb.ipynb

What did you do?

1) I updated my tree diagram with a more accurate definition of a "successful call" according to Cynthia by incorporating dead-ends as successful as well.
2) I performed cluster analysis on when callers are choosing an option for those that enter the staff directory to determine the redundancy of Key 2 and 6 in the Main Menu that both guide callers that want to connect to a staff member. I did a separate analysis for English and Spanish users and plotted the distribution of when callers are making a choice.
3) I analyzed the distribution of the next activity for those entering the HelpWithLegalorOtherReasonMenu to determine if Key 3 and 8 are redundant.

How does it help the project?

1) The updated tree diagram serves as a more accurate analysis on how successful calls moving on from Main Menu are. The increased percentage of successful calls with the new definition of a "successful call" further reinforces the robustness of the Main Menu as a key starting point of the call.
2) Although not perfect, the cluster analysis gave a sense of what keys callers were actually pressing to determine whether or not Key 2 and 6 are truly redundant. This analysis revealed that the majority of callers were not even hearing key 6, suggesting that removing the option and simplifying the Main Menu prompt would be better. In summary, this analysis helped redesign the Main Menu to be more concise.
3) Although the results were not as clear as the analysis above, the last analysis helped answer the question whether or not key 3 and 8 were both necessary. Roughly an equal proportion of callers were entering the Legal Menu, Front Desk or returning to the Main Menu. Discussions with Cynthia seemed necessary to decide whether or not key 8 is redundant, but this analysis helped provide the data on how key 8 was functioning in guiding callers.

Issues faced (if any)
N/A

Attempts to resolve issues (if any)
N/A

Issues resolved (if any)
N/A

Next steps
With regards to next steps, Cynthia's feedback on my new findings is most important to determine which areas I should focus on: whether to dive deeper into the recommendations I made or get my recommendations approved and moving forward with a new action item. From what I am observing, action item 3 of potentially removing key 8 looks the most ambiguous so I am planning to come up with additional layers of analysis to help me better determine the necessity of key 8. I am also brainstorming other parts of the Main Menu that I can perform cluster analysis to find out which parts of the prompts people are moving forward or backward (repeating) to first understand the timing of key selections and ultimately come up with better solutions to reduce the time spent at Main Menu.

References
N/A

Presentation 3

Link to Presentation 3 Slides
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/MenuRepetitionsandSuccessfulCallAnalysis_4Nov_WillVinceYiHyun.pdf

Link to Code
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Menu%20redundancy%20analysis/MainMenuSuccessAnalysis_4Nov_YiHyun.ipynb

What did you do?
After talking with Krish and my team, we decided to focus more on the "successful call" variable so I performed a more granular analysis on the main menu.

First, for calls that reached the Main Menu, I split the data into those that were abandoned at Main Menu and those that went further into the call. Only a handful of calls were abandoned at Main Menu and two thirds of them were after repeating the prompt.

Secondly, after finding out that most callers did not hear the full prompt for Main Menu, I removed the "listened to full prompt" and added the "successful call" layer instead. As a result, 75% of the calls reached a queue at some point, and I found out that roughly half of them reached an actual agent.

To further analyze if these successful calls were stemming from, I found out the distribution of next activities after reaching the Main Menu. Roughly 28% of callers went straight into a queue after hearing the Main Menu prompt. Additionally, 58% went to the Legal Menu which has most of the queue paths. This means that the success of almost 90% of the calls start from the Main Menu.

I finally came up with three suggestions that will help maximize Main Menu's role as the starting node to a queue, mainly revolving around the fact that Main Menu has very little confusion and many paths to a queue.

How does it help the project?
My analysis builds upon our redundancy assessment by identifying repetitive or unnecessary paths that could be streamlined. I focused on the Main Menu, which serves as the first complex decision point with multiple queue options and often leads to the Legal Menu, where the majority of calls are directed. The three key observations and recommendations aim to simplify the Main Menu structure and reduce caller time by improving menu efficiency. Ultimately, these suggestions can reallocate the time from redundant options and potential repeated paths to Legal Menu or other queues.

Issues faced (if any)
N/A

Attempts to resolve issues (if any)
N/A

Issues resolved (if any)
N/A

Next steps
It appears that the Main Menu requires minimal further analysis given its high success rate and low return rate. Therefore, we will shift our focus to the Legal Menu. Specifically, our plan is to analyze the most frequently accessed submenus within the Legal Menu to better understand the sources of confusion or misrouting between them. Additionally, we intend to broaden our analysis from the node level (e.g., Main Menu, Legal Menu) to the entire call session level. Examining full call paths—including redundancies and repetitions—will help us identify the most effective and efficient routes that callers take to achieve successful outcomes. I will have to coordinate with Will and Vincent and which areas I can help on.

References
N/A

Presentation 2

Link to Presentation 2 Slides
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/RepeatedProcesses_23Oct_WillVinceYiHyun.pdf

Link to Code
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Menu%20redundancy%20analysis/MainMenuRedundancies_23Oct_YiHyun.ipynb

What did you do?

Our group (Will, Vincent, Yi Hyun) recategorized the two types of redundancies that we defined in the last presentation into types of people with four subtypes. We've brainstormed the assumptions and algorithms that will divide the CAR data into these four types.

Once we agreed on the definitions, I first analyzed the Main Menu to identify the proportion of each type of caller. The results showed that the majority of people did not listen to the full prompt but were still able to navigate through the call without coming back to the main menu. My second analysis showed the distribution of the number of visits to the MainMenu prompt and found out the majority of calls did not have the caller returning back to the Main Menu. Finally, I focused on call sessions when the caller did have to return back to the Main Menu and found what menu they were coming from and the proportion of those. One significant finding was that the majority of callers proceeded to the LegalMenu2 after the MainMenu, but rarely had to come back to the MainMenu. This is an indication that the MainMenu prompt is concise and efficient enough for callers to jump from MainMenu to LegalMenu2.

How does it help the project?

These analyses are an extension of our definition of redundancies and help our objective of finding possible redundancies within the caller experience and if yes, why that is the case and what segments of the call the redundancies are coming from. In particular, my analyses this time helped our group shift our focus away from the MainMenu since it appeared to be less of a problem of confusion, both path and prompt wise.

Issues faced (if any)
Our main issue was that the CAR data did not tell us if the call was in English or Spanish. We've noticed that the prompt length of English and Spanish was quite different (10+ sec difference), usually Spanish prompts being longer. This made our time analysis on determining whether or not the caller listened to the full option not as accurate since we had to choose a duration that will roughly cover both durations. For example, for the Main Menu prompt analysis, the duration of the English prompt is 52 seconds while the Spanish prompt is 70 seconds so I had to consider all callers exceeding a call duration over 52 seconds to be categorized into "fully

listened to”. This unfortunately includes Spanish callers that did not listen to the full Spanish prompt which can affect our time analysis results.

Attempts to resolve issues (if any)
There was no way as of right now from the CAR data to find the language of the call.
Issues resolved (if any)
N/A

Next steps
Main Menu had a high percentage of people not listening to the full option and never returning back, which is a pretty good indication that Main Menu is not the primary segment causing confusion. Therefore, after finding out if multiple repetitions of Main Menu help callers navigate to a queue, we are planning to shift our focus on Legal Menu which seemed to have more confusion from the higher percentage of people returning back. In addition, there are much more sub-menus branching out for Legal Menu where we can analyze more granularly which paths are most frequent and which prompts might be confusing.

If the language is solvable, we also hope to analyze at which point of time during the prompt, people are leaving to analyze the impact of the duration of listening to a prompt has on the success rate of the caller reaching the right menu and also eventually reaching a queue. As of right now, we were only able to analyze at a binary level to “fully listened to” and “did not fully listen to”.

References
N/A

Presentation 1

Link to Presentation 1 Slides
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/RedundantPaths_09Oct_VincentWillYiHyun.pdf

What did you do?
Our group (Will, Vincent, Yi Hyun) performed flow chart analysis to brainstorm the different possible types and subtypes of each redundancy, the reasons why these should be defined as redundancies, and finally came up with potential algorithms in which those can be identified.

How does it help the project?
Each type of redundancy and proposed algorithm serves as a solid framework in which we can systematically categorize potential redundancies and eventually code to find segments of the call flow that have the most issues.

Issues faced (if any)
One issue as mentioned in the presentation is the potential discrepancy between the call flow chart and CAR data. This may become an issue since most of the proposed algorithms and subtypes were inspired from the flow chart itself and if the CAR data flows differently from the flow chart, our group may face difficulties backtracking each call path.

Attempts to resolve issues (if any)
We flagged one example of the discrepancy to the professor for confirmation/help.

Issues resolved (if any)
We are still waiting for confirmation/response from the professor

Next steps
My next step will involve focusing on a subtype of Type 2 redundancy to create the code that will replicate our proposed algorithm. This will eventually lead to knowing whether the type of subtype of redundancy is even a problem (observation of frequent occurrence) and if yes, if there are ways to mitigate that.

More specifically, our group is planning to analyze the CAR data to create a network chart or code that will map the flow for each caller id. This will help us find which nodes or edges most callers are stuck at or are being repeated.

The group will be collaborating across all types of redundancy. For example, I will be helping Vince by aggregating time data to find paths that take longer than they should or Will by further analyzing the menu chart for code redundancies.

References: N/A